

Muralidharan Vivekananthan

Backend Software Engineer | Go • AWS • PostgreSQL • REST APIs

✉ contact@muralidev.com | ☎ +91 7708484273 | 📍 Bangalore, India | 🌐 [My Portfolio](#) | 🔗 [My LinkedIn](#)

Summary

Software Engineer with nearly 3 years of experience building and improving backend integrations, automation workflows, data synchronization systems, and cloud-based solutions in production SaaS environments. Experienced in breaking down ambiguous problems, extending existing systems, designing practical solutions, and troubleshooting production issues across Go, AWS, PostgreSQL, REST APIs, and asynchronous workflows. Proven experience improving engineering processes and system performance through reusable tooling, configuration-driven solutions, workflow automation, and targeted optimization.

Experience

KAVA CRM

Remote - Australia

CRM Developer — Backend / Integrations

February 2025 – Present

- Built and maintained a Spreadsheet Comparator using React, Go, and AWS to validate synchronized data between systems. Introduced reusable comparison templates and improved the validation workflow, reducing comparison time from nearly a day or more to a few hours.
- Built a CRM synchronization simulator using React, Go, and AWS to upload and validate client, insurer, and policy data from CSV files, store inputs in S3, interact with the Insight API, and trigger ECS-based CRM synchronization workflows.
- Implemented data-management and testing workflows including automatic CSV module detection, validation, selective/full data deletion, real-time status feedback, and ECS task triggering to support integration testing and troubleshooting.
- Migrated the Microsoft SharePoint integration from TypeScript/AWS to Go/AWS, reducing processing time from more than a minute to under a minute and adapting the implementation for multiple customer deployments with customer-specific requirements.
- Extended an existing configuration-driven integration workflow to support AWS S3-based policy-system integrations alongside existing Insight and WinBeat flows. Implemented processing of CSV files from S3, transforming and synchronizing the data into Zoho CRM.
- Improved deployment and configuration workflows by automating credential and configuration setup, reducing setup time from at least 10 minutes to under 10 seconds.
- Used third-party Terraform/Terragrunt infrastructure templates for customer deployments, executing infrastructure changes and working with infrastructure-as-code workflows.
- Investigated production integration failures systematically using application logs, API responses, data comparisons, and workflow execution details to identify root causes and resolve synchronization and processing issues.
- Analysed existing Zoho CRM integrations and Go/AWS workflows to identify unnecessary processing and infrastructure usage, contributing to cost-reduction efforts across deployed integration solutions.

ZUPER

Chennai, India

Software Engineer Trainee

October 2023 – December 2024

- Developed and maintained backend automation workflows and API integrations connecting Zuper with SaaS platforms including Zoho CRM, Zoho Books, Zoho Desk, Zoho Inventory, Zoho Projects, Freshsales, Freshdesk, Zendesk, and HubSpot.
- Designed and implemented integration workflows using Deluge, Node.js, Express.js, and n8n, handling API communication, data transformation, synchronization, and workflow execution across interconnected systems.
- Analyzed client business requirements and translated them into practical backend integration and automation solutions within existing product architecture.

- Implemented data synchronization workflows between business applications, improving consistency of business data and reducing manual operational processes.
- Investigated integration and automation failures through log analysis, API inspection, data validation, and workflow debugging, identifying root causes and implementing fixes across customer environments.
- Developed and integrated custom UI functionality within Zuper to support customer-specific workflows and business requirements.

Skills

- 🔗 **Languages:** Go, JavaScript, Lua, Deluge, SQL
- 🔗 **Backend:** REST APIs, API Integrations, Webhooks, Event-Driven Architecture, Backend Automation
- 🔗 **Databases:** PostgreSQL
- 🔗 **Frontend:** React
- 🔗 **AWS:** S3, SQS, Lambda, ECS, API Gateway, EventBridge, EC2, RDS, Parameter Store
- 🔗 **DevOps:** Docker, GitHub Actions, Terraform/Terragrunt, CI/CD, AWS CLI
- 🔗 **Engineering:** System Architecture, API Design, Asynchronous Processing, Message Queues, Fault Handling, Root Cause Analysis, Log Analysis, Debugging
- 🔗 **Other:** Zoho Products, Deluge Scripting, VBA Excel, Git, Postman, n8n

Projects

ApplyFlow – Event-Driven Job Application Automation

- [Go • React • PostgreSQL • AWS RDS • S3 • SQS • EventBridge • EC2 • ALB • Docker • GitHub Actions • Gmail API]
- **GitHub Repo Link:** github.com/murali-vivek/applyFlow [Access could be provided on request.]
 - Designed and built an event-driven job outreach platform using Go, PostgreSQL, and AWS to automate personalized email campaigns through the Gmail API.
 - Built asynchronous campaign processing with Amazon SQS and a worker service, decoupling API requests from email delivery and supporting failure recovery and at-least-once processing.
 - Implemented campaign workflows including XLSX validation, duplicate-recipient prevention, email templates, resume attachments, scheduling, cancellation, and outreach progress tracking.
 - Implemented Google OAuth authorization and Gmail API integration, with Amazon S3 for file storage and EventBridge for scheduled campaign processing.
 - Built the React frontend and deployed the application on AWS using Docker, Amazon RDS, EC2, Application Load Balancer, and security groups.

Education

PSNACET	Bachelor of Engineering • 8
Computer Science	Dindigul, India • Aug 2019 - June 2023